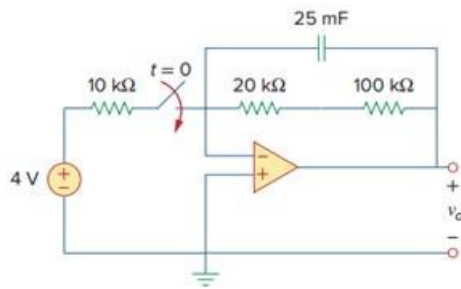


Problem

For the op amp circuit in Fig. 7.134, find $v_o(t)$ for $t > 0$.

Fig. 7.134:



Step-by-step solution

Step 1 of 2

The voltage V_X be the capacitor voltage for $t < 0$,

$$V_X(0) = 0 \text{ V}$$

For $t > 0$, $20 \text{ k}\Omega$, and $100 \text{ k}\Omega$ resistors are in series & together, they are in parallel with capacitor since no current enters op-amp terminals as $t \rightarrow \infty$, the capacitor acts like an open circuit so that,

$$V(\infty) = -\left(\frac{R_f}{R_{in}}\right)V_i$$

$$\begin{aligned} V(\infty) &= \frac{-4}{10 \text{ k}\Omega} (20 \text{ k}\Omega + 100 \text{ k}\Omega) \\ &= -48 \end{aligned}$$

Calculate the value of R_{th} .

$$\begin{aligned} R_{th} &= 20 + 100 \\ &= 120 \text{ k}\Omega \end{aligned}$$

Calculate the time constant.

$$\begin{aligned} \tau &= R_{th}C \\ &= (120 \times 10^3)(25 \times 10^{-3}) \\ &= 3000 \end{aligned}$$

Step 2 of 2

Hence, the output voltage is,

$$\begin{aligned}V_0(t) &= V_0(\infty) + [V_0(0) - V_0(\infty)]e^{\frac{-t}{\tau}} \\&= -48 + [0 - (-48)]e^{\frac{-t}{3000}} \\&= -48 + 48e^{\frac{-t}{3000}} \\&= -48\left(1 - e^{\frac{-t}{3000}}\right) \text{ V} \\&= 48\left(e^{\frac{-t}{3000}} - 1\right) \text{ V}\end{aligned}$$

Therefore, the output voltage is $\boxed{48\left(e^{\frac{-t}{3000}} - 1\right) \text{ V}}$.